

Maryam Amjad

Machine Learning / AI Engineer | LLM & RAG | Deep Learning | Computer Vision
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SUMMARY

AI/ML Engineer with production experience across crypto, data-intelligence, and healthcare AI. Built and shipped end-to-end systems: a real-time crypto prediction pipeline, a knowledge-graph-backed RAG engine, and a multi-modal clinical AI assistant. Core strength: taking a system from raw data to something people actually use.

EXPERIENCE

Junior Machine Learning Engineer

Tensor Labs

Sep 2025 – Present

Built and shipped production ML/LLM systems across crypto, data-intelligence, and healthcare products; own data pipelines, model integration, and reliability.

Real-Time Crypto Breakout Prediction System

Problem: Traders needed early breakout signals across multiple volatile crypto markets from continuous streaming data.

Role: Designed and deployed the real-time prediction pipeline end-to-end.

Tech: Python, Pandas, Scikit-learn, PostgreSQL, WebSockets, Streamlit.

Outcome: Live system processing 5-minute streaming data for ETH, DOGE, and SOL, serving breakout alerts and a liquidation/liquidity-risk heatmap through a real-time dashboard.

RAG-Based Text-to-SQL Engine (DiscoverIQ)

Problem: Non-technical users needed to query large relational databases without writing SQL, but LLM-generated SQL often failed on real schemas on the first try.

Role: Built the self-correcting SQL feedback loop: when a generated query fails, the agent analyzes the error, asks the model for a corrected query, validates syntax before re-running it, and retries with duplicate-query detection.

Tech: Python, OpenAI, Groq (Llama-3.3-70B), JSON repair/validation, PostgreSQL.

Outcome: Automatically recovers from failed SQL generations without user intervention, using error-analysis prompting, syntax pre-validation, and duplicate-query detection to converge on a working query.

Multi-Modal Clinical AI Assistant (MedGemma)

Problem: Clinicians needed one assistant for image diagnostics, prescription reading, and drug-safety checks instead of juggling separate tools.

Role: Built core backend modules integrating four specialized models.

Tech: MedGemma-4B, TxGemma-9B, Qwen3-VL, Whisper, FastAPI, PyDICOM, FDA/RxNorm APIs.

Outcome: Working assistant combining medical image/DICOM diagnostics, prescription OCR, and live FDA/RxNorm-grounded drug-interaction checks behind one API.

Artificial Intelligence Intern

AIM Lab, Islamabad

Jun 2023 – Aug 2023

Built a tool that auto-generates presentation slide decks from natural-language prompts using NLP and computer vision; applied pre-trained vision models for content understanding and image matching.

TECHNICAL SIDE PROJECTS

LLM Evaluation Toolkit: Open-source toolkit covering 12 LLM evaluation categories (text generation, summarization, RAG, text-to-SQL, safety & robustness, long-context reasoning, multi-modal) with runnable notebooks and an interactive Streamlit dashboard; used as the basis for an internal engineering workshop. *Python, Groq, RAGAS, DeepEval, HuggingFace Evaluate, PromptBench.*

TherapEase (Autism Therapy Assistant): 3D digital twin with real-time facial emotion detection to support autism therapy sessions, combining a browser-based 3D avatar with a live computer-vision emotion-classification backend (final year project). *React, Three.js, OpenCV, MediaPipe, Flask,*

TensorFlow, DeepFace.

Roman Urdu Chatbot: Rasa-based conversational AI for Roman Urdu (code-mixed transliterated Urdu) small talk, with 30+ intents, rule-based dialogue policies, and custom Python actions for dynamic responses; extended with a full voice-to-voice mode using Google Speech-to-Text and gTTS. Deployed live on Streamlit Cloud. *Rasa, Streamlit, gTTS, SpeechRecognition, Python.*

SKILLS

ML/AI: PyTorch, TensorFlow, Keras, Scikit-learn, LangChain, Hugging Face Transformers, RAG, Prompt Engineering, Agentic Workflows, RAGAS, DeepEval

Domains: Computer Vision, NLP, LLMs/Agents, Knowledge Graphs, Time Series, OCR/Speech (Whisper, Mem0)

Data: Neo4j, Qdrant, PostgreSQL, MySQL, SQLAlchemy, Pandas, NumPy, OpenCV, NLTK, SpaCy

Cloud/DevOps: AWS, Docker, FastAPI, Flask, CI/CD, Git

EDUCATION

B.S. in Artificial Intelligence

FAST NUCES, Islamabad

Coursework: Deep Learning, Generative AI, Computer Vision, NLP

Jul 2021 – Jun 2025

CERTIFICATIONS

AWS Certified AI Practitioner (2026) • Generative AI with LLMs, DeepLearning.AI (2025) • AI Agents with RAG and LangChain, IBM (2025) • Convolutional Neural Networks, DeepLearning.AI (2024) • AWS Cloud Essentials (2025)